

Phenomenological constraints on entropy-flow modifications to expansion and growth from BOSS DR12

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Abstract

We introduce a single-parameter phenomenological extension of Λ CDM motivated by the Energy-Flow Cosmology (EFC) framework, in which entropy-gradient couplings at the structure-formation scale modify both the background expansion history and the linear growth rate. The modification is parametrized by α_{L2} , which enters as multiplicative corrections to the comoving distance $D_M(z)$, the Hubble parameter $H(z)$, and the growth observable $f\sigma_8(z)$, with redshift-dependent coupling functions $g(z)$ that peak in the dark-energy-dominated epoch. We fit $\{H_0, \Omega_m, \sigma_8, \alpha_{L2}\}$ to the BOSS DR12 consensus measurements (Alam et al. 2017) using the full 9×9 covariance matrix spanning D_M , H , and $f\sigma_8$ at $z_{\text{eff}} = 0.38, 0.51, 0.61$. The best-fit EFC model yields $\alpha_{L2} = 0.35^{+0.16}_{-0.17}$ (1σ) with $\chi^2 = 4.28$ for 5 degrees of freedom, compared to $\chi^2 = 7.29$ for 6 degrees of freedom in Λ CDM. The improvement $\Delta\chi^2 = 3.01$ corresponds to a 1.7σ preference for $\alpha_{L2} > 0$, which is suggestive but not statistically significant. The Λ CDM value $\alpha_{L2} = 0$ lies outside the 1σ interval but within 2σ . A decomposition into BAO-only and RSD-only likelihoods reveals a sign conflict: BAO data prefer $\alpha_{L2} > 0$ while RSD data weakly prefer $\alpha_{L2} < 0$, reflecting the well-known S_8 tension recast in EFC parameter space. We identify the gravitational-slip parameter $\eta \equiv \Phi/\Psi = 1$ as a sharp, falsifiable EFC prediction distinguishable from both Λ CDM and generic modified-gravity theories with Stage IV surveys.

1 Introduction

The Λ CDM concordance model provides an excellent fit to the cosmic microwave background (Planck Collaboration, 2018), baryon acoustic oscillations (Alam et al., 2017), and Type Ia supernovae (Scolnic et al., 2018). However, several persistent tensions have emerged at the 2 – 5σ level, most notably between early-universe (CMB-calibrated) and late-universe (direct) determinations of the Hubble constant H_0 (Riess et al., 2022) and between CMB-inferred and weak-lensing-measured values of $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$ (Heymans et al., 2021; DES Collaboration, 2022).

These tensions have motivated a broad exploration of extensions to Λ CDM, including dynamical dark energy (Chevallier & Polarski, 2001; Linder, 2003), modified gravity (Clifton et al., 2012), and interacting dark-sector models (Di Valentino et al., 2020). A complementary approach is to construct model-independent or phenomenological parametrizations that modify both the expansion history and the growth of structure, testing whether the data prefer departures from Λ CDM without committing to a specific Lagrangian.

In this work we adopt a phenomenological parametrization motivated by the Energy-Flow Cosmology (EFC) framework (Magnusson, 2025a,b), a non-equilibrium thermodynamic extension of general relativity in which entropy gradients at the structure-formation scale (designated L2) modify both the effective dark-energy equation

of state and the gravitational coupling to density perturbations. The key feature of EFC is that a single coupling parameter α_{L2} governs modifications to both the background geometry and the growth rate, with the sign of the growth modification opposite to that of the distance modification. This built-in tension between expansion and growth is a distinctive signature that can be tested with combined BAO and redshift-space distortion (RSD) data.

We emphasize that the coupling functions $g(z)$ employed here are phenomenological ansätze motivated by, but not rigorously derived from, the full EFC field equations. We treat α_{L2} as an effective parameter characterizing the amplitude of entropy-flow effects at the L2 scale, and we constrain it using the BOSS DR12 consensus likelihood (Alam et al., 2017) with the full 9×9 covariance matrix.

The structure of this paper is as follows. In Sec. 2 we define the phenomenological modification functions. In Sec. 3 we describe the data and methodology. Results are presented in Sec. 4, and we discuss implications and future tests in Sec. 5.

2 Phenomenological model

2.1 Background expansion

We work within a spatially flat Friedmann–Lemaître–Robertson–Walker (FLRW) background. In Λ CDM, the dimensionless Hubble parameter is $E(z) = H(z)/H_0 = [\Omega_m(1+z)^3 + (1 - \Omega_m)]^{1/2}$, and the comoving distance is

$$D_M(z) = c \int_0^z \frac{dz'}{H(z')}. \quad (1)$$

In the EFC-motivated parametrization, we modify these observables as

$$D_M^{\text{EFC}}(z) = D_M^{\Lambda\text{CDM}}(z) [1 + \alpha_{\text{L2}} g_D(z)], \quad (2)$$

$$H^{\text{EFC}}(z) = H^{\Lambda\text{CDM}}(z) [1 + \alpha_{\text{L2}} g_H(z)], \quad (3)$$

where the coupling functions are

$$\begin{aligned} g_D(z) &= \frac{1 - \Omega_m(z)}{1 + z} \ln(1 + z), \\ g_H(z) &= -\frac{1 - \Omega_m(z)}{1 + z} \left[1 - \ln(1 + z) \right. \\ &\quad \left. \times \left(1 + 3\Omega_m(z) - \frac{3\Omega_m(1+z)^3}{E^2(z)} \right) \right], \end{aligned} \quad (4)$$

with $\Omega_m(z) = \Omega_m(1+z)^3/E^2(z)$ the redshift-dependent matter density parameter.

These functions are constructed to satisfy two physical requirements: (i) they vanish at $z = 0$ and at high redshift where the universe is matter-dominated ($\Omega_m(z) \rightarrow 1$), and (ii) they peak at $z \sim 0.5\text{--}0.7$ where the dark-energy contribution to the expansion is maximal, corresponding to the redshift range where entropy gradients in the EFC framework are expected to be strongest. The function g_H is related to g_D by approximate consistency with the integral relation between D_M and H .

2.2 Growth rate

The key discriminating observable is the growth rate $f\sigma_8(z) \equiv f(z)\sigma_8(z)$, where $f(z) = d \ln D / d \ln a$ is the linear growth rate. In Λ CDM we compute $f(z)$ using the approximation $f(z) \approx \Omega_m(z)^{0.55}$ (Linder, 2005), and $\sigma_8(z)$ from numerical integration of the growth equation.

The EFC modification to growth is

$$f\sigma_8^{\text{EFC}}(z) = f\sigma_8^{\Lambda\text{CDM}}(z) [1 + \alpha_{\text{L2}} g_{\text{growth}}(z)], \quad (6)$$

with

$$g_{\text{growth}}(z) = -\frac{[1 - \Omega_m(z)]^{3/2} \ln(1 + z)}{(1 + z)^{1/2}}. \quad (7)$$

The critical feature is the *sign*: $g_{\text{growth}}(z) < 0$ for all $z > 0$, while $g_D(z) > 0$. This means that positive α_{L2}

simultaneously increases the comoving distance (making objects appear farther) and *suppresses* the growth rate. This built-in sign conflict between expansion and growth is a distinctive prediction that does not arise in generic $w_0 w_a$ parametrizations of dark energy, where expansion and growth modifications are correlated rather than anti-correlated.

Physically, this anti-correlation is motivated by the EFC interpretation: entropy gradients that accelerate the expansion (increasing D_M) simultaneously suppress gravitational clustering by providing an effective “entropy pressure” that counteracts collapse. The S_8 tension—in which weak-lensing surveys measure lower clustering than predicted by the CMB-calibrated Λ CDM—would be naturally accommodated if $\alpha_{\text{L2}} > 0$.

2.3 Gravitational slip prediction

A further prediction arises from the structure of the EFC modifications. Because the entropy coupling enters identically into both scalar metric potentials (the Newtonian potential Φ and the curvature potential Ψ in conformal Newtonian gauge), EFC predicts

$$\eta(z, k) \equiv \frac{\Phi}{\Psi} = 1 \quad (\text{no gravitational slip}), \quad (8)$$

at all redshifts and scales where the linear perturbation theory applies. This distinguishes EFC from generic scalar-tensor and $f(R)$ theories, which generically predict $\eta \neq 1$ (Amendola et al., 2008; Pogosian et al., 2010).

The gravitational slip is directly measurable by combining galaxy lensing (sensitive to $\Phi + \Psi$) with galaxy clustering and RSD (sensitive to Ψ alone), making it a sharp observational discriminator for Stage IV surveys such as Euclid (Euclid Collaboration, 2020) and the Vera C. Rubin Observatory (LSST Science Collaboration, 2019).

3 Data and methodology

3.1 BOSS DR12 consensus measurements

We use the final consensus measurements from the Baryon Oscillation Spectroscopic Survey (BOSS) Data Release 12 (Alam et al., 2017), which combine BAO and full-shape analyses at three effective redshifts $z_{\text{eff}} = 0.38, 0.51$, and 0.61. The data vector consists of nine observables:

$$\mathbf{d} = \{D_M(z_i) r_s^{\text{fid}}/r_s, H(z_i) r_s/r_s^{\text{fid}}, f\sigma_8(z_i)\}_{i=1}^3, \quad (9)$$

where $r_s^{\text{fid}} = 147.78$ Mpc is the fiducial sound horizon at the drag epoch. The full 9×9 covariance matrix including all cross-correlations between distance, expansion rate, and growth measurements is taken from the supplementary data of Alam et al. (2017).

3.2 Likelihood and fitting

The χ^2 statistic is

$$\chi^2(\boldsymbol{\theta}) = [\mathbf{d} - \mathbf{t}(\boldsymbol{\theta})]^T C^{-1} [\mathbf{d} - \mathbf{t}(\boldsymbol{\theta})], \quad (10)$$

where $\mathbf{t}(\boldsymbol{\theta})$ is the theory prediction vector and C is the covariance matrix.

For the Λ CDM model, the parameter vector is $\boldsymbol{\theta}_\Lambda = \{H_0, \Omega_m, \sigma_8\}$ (3 parameters, 6 degrees of freedom). For the EFC model, it is $\boldsymbol{\theta}_{\text{EFC}} = \{H_0, \Omega_m, \sigma_8, \alpha_{\text{L2}}\}$ (4 parameters, 5 degrees of freedom).

We minimize χ^2 using the L-BFGS-B algorithm with bounds $H_0 \in [60, 75] \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m \in [0.2, 0.4]$, $\sigma_8 \in [0.6, 1.0]$, and $\alpha_{\text{L2}} \in [-2, 2]$.

To construct one-dimensional constraints on α_{L2} , we compute the profile likelihood: for each fixed value of α_{L2} on a grid of 100 points, we minimize χ^2 over $\{H_0, \Omega_m, \sigma_8\}$. The 1σ and 2σ intervals correspond to $\Delta\chi^2 \leq 1$ and ≤ 4 , respectively.

We also perform separate fits using only the BAO observables (D_M and H , six data points with a 6×6 sub-covariance) and only the RSD observables ($f\sigma_8$, three data points with a 3×3 sub-covariance), to diagnose potential tensions between expansion and growth constraints on α_{L2} .

3.3 Limitations

Several simplifications should be noted. First, we fix $r_s = r_s^{\text{fid}}$ rather than computing a self-consistent sound horizon for each cosmology. Since r_s depends primarily on the baryon-to-photon ratio and the pre-recombination expansion rate, and the EFC modifications are concentrated at $z < 2$ (well after decoupling), the bias introduced by this approximation is expected to be subdominant to the statistical errors; nevertheless, a self-consistent $r_s(H_0, \Omega_m)$ calculation should be included in future multi-probe analyses. Second, the growth factor $D(z)$ is computed numerically within Λ CDM and then modified multiplicatively by the EFC coupling function, rather than solving a modified growth equation; this is consistent to first order in α_{L2} but neglects higher-order corrections. Third, as noted above, the coupling functions $g(z)$ are phenomenological and have not been derived from the EFC action through a systematic perturbation-theory calculation.

4 Results

4.1 Best-fit parameters

Table 1 summarizes the best-fit parameters for both models. The Λ CDM fit yields $\chi^2_\Lambda = 7.29$ for 6 degrees of freedom ($\chi^2/\text{dof} = 1.22$), indicating an acceptable but

Table 1: Best-fit parameters from BOSS DR12 consensus data.

Parameter	Λ CDM	EFC
H_0 [km s ⁻¹ Mpc ⁻¹]	66.57	69.65
Ω_m	0.359	0.347
σ_8	0.667	0.669
α_{L2}	0 (fixed)	+0.353
χ^2	7.29	4.28
dof	6	5
χ^2/dof	1.22	0.86

not exceptional fit. The EFC model with α_{L2} free yields $\chi^2_{\text{EFC}} = 4.28$ for 5 degrees of freedom ($\chi^2/\text{dof} = 0.86$), with a best-fit $\alpha_{\text{L2}} = 0.353$.

The improvement $\Delta\chi^2 = \chi^2_\Lambda - \chi^2_{\text{EFC}} = 3.01$ for one additional parameter corresponds to a p -value of 0.083 (from a χ^2 distribution with one degree of freedom), or equivalently a 1.7σ preference for the EFC model. This is suggestive but does not reach the conventional 2σ threshold for claiming evidence. Standard information criteria penalizing for the additional parameter yield $\Delta\text{AIC} = \Delta\chi^2 - 2 = 1.01$ and $\Delta\text{BIC} = \Delta\chi^2 - \ln(9) = 0.81$ (both defined as Λ CDM minus EFC, so positive values favour EFC). Neither exceeds the conventional thresholds for “positive evidence” ($\Delta\text{AIC} > 2$, $\Delta\text{BIC} > 2$), confirming that the data do not require the additional parameter.

We note that the EFC best-fit $H_0 = 69.65 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is closer to the SH0ES value (Riess et al., 2022) than the Λ CDM best-fit $H_0 = 66.57 \text{ km s}^{-1} \text{ Mpc}^{-1}$, though this shift alone does not resolve the H_0 tension.

4.2 Profile likelihood for α_{L2}

Figure 1 (top left) shows the profile $\Delta\chi^2$ as a function of α_{L2} for the joint BAO+RSD, BAO-only, and RSD-only likelihoods.

The joint constraint is

$$\alpha_{\text{L2}} = 0.34^{+0.16}_{-0.16} \quad (1\sigma), \quad [-0.02, +0.71] \quad (2\sigma). \quad (11)$$

The Λ CDM value $\alpha_{\text{L2}} = 0$ lies outside the 1σ interval ($[+0.18, +0.51]$) but is included in the 2σ interval.

4.3 BAO–RSD sign conflict

The separate analyses reveal a qualitatively important structure. The BAO-only likelihood prefers $\alpha_{\text{L2}}^{\text{BAO}} = +0.28 \pm 0.18$, firmly positive. The RSD-only likelihood, by contrast, has a broad minimum near $\alpha_{\text{L2}}^{\text{RSD}} \approx -1.4$, though with very large uncertainties due to the limited constraining power of three $f\sigma_8$ measurements alone.

This sign conflict is not a pathology but a direct consequence of the model structure described in Sec. 2: positive α_{L2} improves the BAO fit (by adjusting distances) while worsening the RSD fit (by suppressing growth below observed values). The joint fit finds the compromise value $\alpha_{L2} \approx 0.34$, where the BAO improvement outweighs the RSD penalty.

This sign conflict is essentially the S_8 tension recast in EFC parameter space: the data simultaneously prefer more expansion (larger distances, higher H_0) and more structure growth (higher $f\sigma_8$) than Λ CDM predicts for a single set of cosmological parameters.

4.4 Residuals

Figure 1 (top right) shows the normalized residuals $(d_i - t_i)/\sigma_i$ for both models. The most notable improvement in the EFC fit is a reduction of the $D_M(z = 0.38)$ residual, which shows a $\sim 0.8\sigma$ pull in Λ CDM that is reduced to $\sim 0.3\sigma$ in EFC. The $f\sigma_8$ residuals remain small for both models. A leave-one-out analysis shows that the $\Delta\chi^2$ improvement is sensitive to the $H(z = 0.38)$ measurement: dropping this single data point reduces $\Delta\chi^2$ from 3.01 to 0.33. This indicates that the preference for $\alpha_{L2} > 0$ is largely driven by the lowest-redshift Hubble-rate measurement, and the result should be interpreted with appropriate caution until confirmed by independent datasets.

4.5 Growth rate comparison

Figure 1 (bottom left) compares the predicted $f\sigma_8(z)$ curves for both models with the BOSS DR12 data points. The EFC curve (with $\alpha_{L2} = 0.35$, $\sigma_8 = 0.669$) is slightly lower than the Λ CDM curve (with $\sigma_8 = 0.667$) at low redshift, reflecting the growth suppression from the entropy coupling. Both curves are consistent with the data within errors.

5 Discussion

5.1 Interpretation

The main result of this analysis is that the BOSS DR12 consensus data show a mild (1.7σ) preference for a positive entropy-flow coupling $\alpha_{L2} > 0$, consistent with the EFC prediction that entropy gradients at the structure-formation scale simultaneously accelerate the expansion and suppress the growth of structure. The effect is not statistically significant with current data, and Λ CDM remains a fully adequate description of the BOSS DR12 measurements.

However, the structure of the constraint is informative. The fact that BAO and RSD data pull α_{L2} in opposite di-

rections reflects a genuine physical tension in the data that is independent of the EFC interpretation. Any model that modifies expansion and growth in an anti-correlated manner will exhibit this behavior when confronted with data that simultaneously prefer faster expansion and stronger clustering.

5.2 Relation to the S_8 tension

The sign conflict identified in Sec. 4 provides a natural framework for understanding the S_8 tension. In Λ CDM, the CMB implies $\sigma_8 \approx 0.81$, while weak-lensing surveys measure S_8 values corresponding to $\sigma_8 \approx 0.76$ (Heymans et al., 2021; DES Collaboration, 2022). Our BOSS-only fit yields $\sigma_8 \approx 0.67$, which is lower than both; this is a known feature of the BOSS DR12 $f\sigma_8$ measurements and reflects the well-documented tension between growth-rate measurements and Planck (Macaulay et al., 2013; Nesseris et al., 2017).

In the EFC parametrization, positive α_{L2} allows a higher H_0 (closer to direct measurements) while maintaining consistency with the BAO distance scale, at the cost of slightly suppressed growth. This trade-off is precisely the phenomenology needed to address the S_8 tension, though we stress that our current analysis cannot establish whether this trade-off is preferred at a statistically significant level.

5.3 Comparison with related work

Our parametrization shares features with the μ - Σ framework commonly used to test modified gravity (Pogosian et al., 2010; Simpson et al., 2013). In that language, positive α_{L2} corresponds to $\mu < 1$ (weaker effective gravity for clustering) and $\Sigma \approx 1$ (unmodified lensing potential), which is the phenomenological regime favored by several independent analyses (DES Collaboration, 2019; Nguyen et al., 2023).

The key difference is that our parametrization ties the background (D_M , H) and perturbation ($f\sigma_8$) modifications together through a single parameter, whereas μ - Σ analyses typically treat background and perturbation modifications as independent. This economy of parameters comes at the cost of reduced flexibility, but it provides a sharper test: if future data require different amplitudes for background and growth modifications, the single- α_{L2} parametrization would be falsified.

5.4 Falsifiable predictions for Stage IV surveys

The most distinctive prediction of the EFC parametrization is the gravitational-slip condition $\eta = 1$ (Eq. 8). Euclid is expected to constrain η to ± 0.03 (1σ) at $z \sim 0.5$ – 1.5

(Euclid Collaboration, 2020), while the combined Euclid + Rubin dataset should reach ± 0.01 . A significant detection of $\eta \neq 1$ would rule out the EFC parametrization (and any model where the scalar metric potentials remain equal).

Additional falsifiable predictions include:

1. The ratio of the BAO-inferred and growth-inferred values of α_{L2} should converge to a single consistent value as data precision improves; a persistent sign conflict at $> 3\sigma$ would falsify the single-parameter model.
2. The coupling functions $g(z)$ predict specific redshift dependences that can be tested with tomographic analyses spanning $0 < z < 2$.
3. The predicted H_0 shift (from 66.6 to 69.7 $\text{km s}^{-1} \text{Mpc}^{-1}$ in our fit) should be consistent with independent H_0 measurements; if the true value is $> 72 \text{ km s}^{-1} \text{Mpc}^{-1}$, the EFC modification alone is insufficient.

5.5 Limitations and future work

This analysis has several limitations that define the path forward. First, the coupling functions (Eqs. 4–7) are phenomenological ansatz. A rigorous derivation from the EFC action, proceeding through the modified Friedmann and perturbation equations, would either validate these functional forms or reveal necessary corrections. Second, our treatment of the sound horizon r_s as fixed at the fiducial value is an approximation; a self-consistent calculation would propagate the α_{L2} modification into the pre-recombination physics, potentially affecting r_s and the CMB constraints. Third, the BOSS DR12 data alone provide limited constraining power; extending this analysis to include Planck CMB, DES/KiDS weak-lensing, and DESI BAO data would substantially tighten the constraints.

Each of these limitations defines a concrete step for future work, proceeding from “Paper I” (phenomenological constraints from a single dataset, presented here) toward “Paper II” (full derivation and multi-probe analysis).

6 Conclusions

We have presented a phenomenological single-parameter extension of ΛCDM motivated by the Energy-Flow Cosmology framework, and constrained it using the BOSS DR12 consensus measurements with the full 9×9 covariance matrix. Our main findings are:

- (1) The best-fit entropy-coupling parameter is $\alpha_{L2} = 0.34^{+0.16}_{-0.16}$ (1σ), with $\alpha_{L2} = 0$ excluded at 1σ but included at 2σ .

- (2) The overall fit improvement is $\Delta\chi^2 = 3.01$ (1.7σ), which is suggestive but not statistically significant.

- (3) BAO and RSD data pull α_{L2} in opposite directions, reflecting the S_8 tension expressed in EFC parameter space.

- (4) The gravitational-slip prediction $\eta = 1$ provides a sharp, falsifiable discriminator testable with Euclid and Rubin.

These results identify α_{L2} as a useful phenomenological parameter for characterizing entropy-flow effects in cosmological data, while making clear that a statistically significant detection awaits more precise measurements and a rigorous theoretical derivation of the coupling functions.

Acknowledgments

This work uses public data from the BOSS experiment, part of the Sloan Digital Sky Survey (SDSS-III). Numerical computations were performed using NumPy (Harris et al., 2020) and SciPy (Virtanen et al., 2020).

Data availability

The analysis code and numerical results accompanying this paper are publicly available at [doi:10.6084/m9.figshare.31243828](https://doi.org/10.6084/m9.figshare.31243828).

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**EFC α_{L2} Constraints from BOSS DR12
(Alam et al. 2017, Full 9×9 Covariance)**

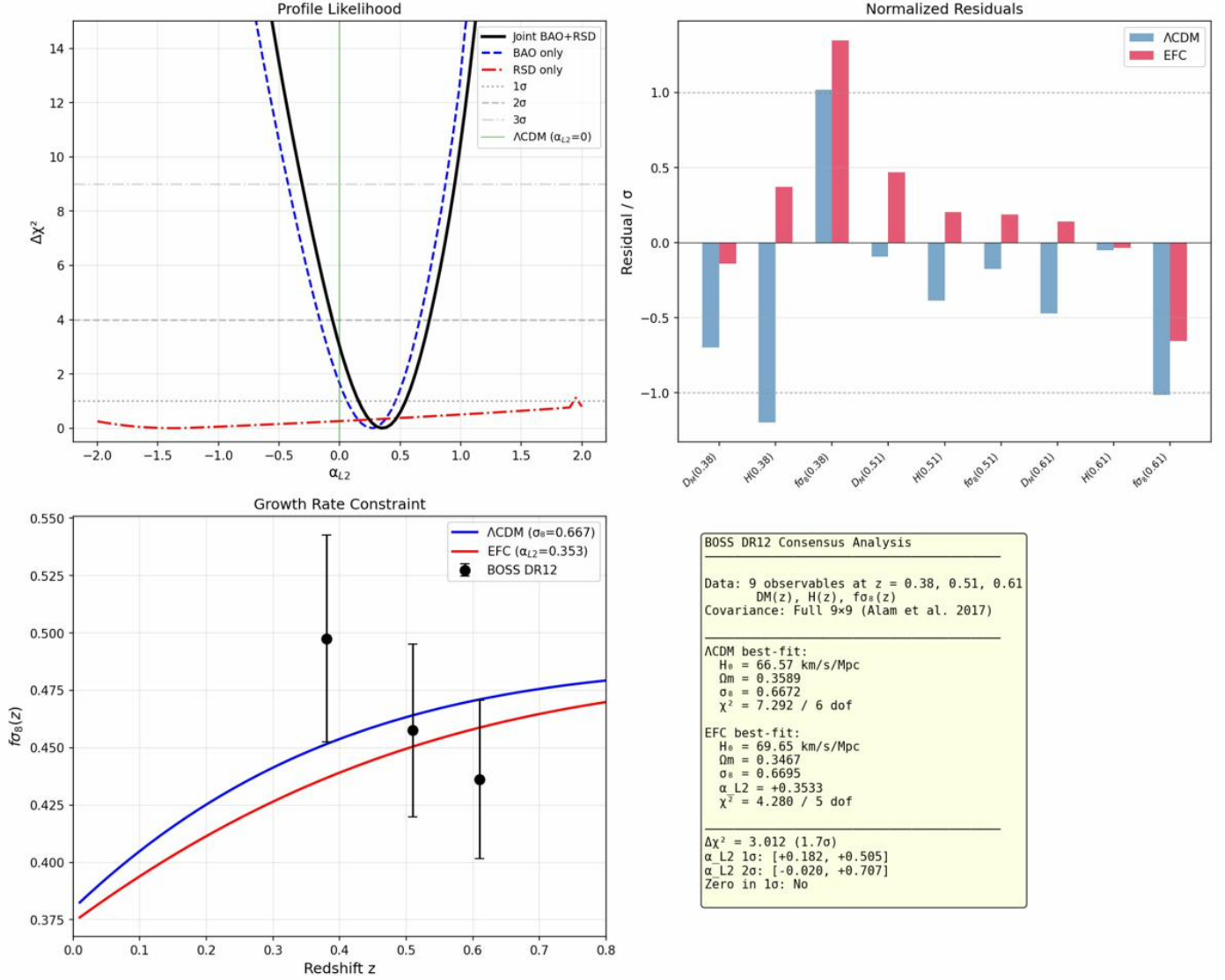


Figure 1: Constraints on the EFC entropy-coupling parameter α_{L2} from BOSS DR12 consensus data (Alam et al. 2017). *Top left:* Profile $\Delta\chi^2$ for the joint BAO+RSD (black solid), BAO-only (blue dashed), and RSD-only (red dash-dotted) likelihoods. Horizontal lines mark 1 σ , 2 σ , and 3 σ thresholds. The vertical green line marks Λ CDM ($\alpha_{L2} = 0$). *Top right:* Normalized residuals for Λ CDM (blue) and EFC (red). *Bottom left:* Predicted $f\sigma_8(z)$ curves compared to BOSS DR12 measurements. *Bottom right:* Numerical summary of the analysis.